

Sarang Gosavi

🏠 Boulder, CO, USA | 📞 +1 (475) 619-0266
✉️ sarang.gosavi@colorado.edu | 🐙 github | 🔗 linkedin | 🌐 ORCID

Education

University of Colorado, Boulder

Start date August 2026

Ph.D. Paul M. Rady Department of Mechanical Engineering

University Of Colorado, Boulder

August 2024 - expected completion May 2027

M.S. (Thesis) Paul M. Rady Department of Mechanical Engineering

Savitribai Phule Pune University

August 2014 - August 2019

B.E. in Mechanical Engineering

Awards and Honours

Dwight E. and Jessie D. Ryland Graduate Fellowship

August 2026

College of Engineering and Applied Science at CU Boulder

CEAS I&E Ascent Fellowship

June 2026

Venture Partners at CU Boulder

Lab Venture Challenge (LVC 2025)

October 2025

Colorado, Office of Economic Development and International Trade (OEDIT), Venture Partners CU Boulder.

NSF National I-Corps

June 2025

Product we need Award

Fall 2025 Outstanding MS Mechanical Engineering Course Assistant Award

November 2025

Paul M. Rady Department of Mechanical Engineering, University of Colorado, Boulder.

Mechanical Engineering MS Support Award (WEFTEC 2025)

October 2025

Dana Shinn and Paul M. Rady Department of Mechanical Engineering, University of Colorado, Boulder.

Mechanical Engineering Travel Award (Rocky Mountain Water Conference (RMWC))

August 2025

Paul M. Rady Department of Mechanical Engineering, University of Colorado, Boulder.

Mechanical Engineering Graduate Scholarship

August 2025, 2024, January 2025

Paul M. Rady Department of Mechanical Engineering, University of Colorado, Boulder.

ME MS Departmental Scholarship

August 2024

Paul M. Rady Department of Mechanical Engineering, University of Colorado, Boulder.

Patents

On-chip Catalytic System

Oct 2025

Provisional Patent Application No. 63/902,813

Research Experience

Plasmonic Photocatalysis

August 2024 - present

Research Assistant, Department of Mechanical Engineering, University of Colorado, Boulder

Advisor: Prof. Longji Cui, Department of Mechanical Engineering, University of Colorado, Boulder

- Synthesized plasmonic nanoparticle catalysts for thermo-photocatalytic applications.
- Designed and developed a custom catalyst-reactor system enabling synergistic thermal and photonic activation for enhanced catalytic performance.
- Performed quantitative gas-phase analysis using gas chromatography, including calibration, sampling, and peak-integration.
- Demonstrated complete ammonia decomposition using the newly developed plasmonic catalyst-reactor system, as a proof-of-concept for H₂ production.

Quantum Algorithmic Statistical Mechanics (QASM)

August 2022 - July 2024 (part-time)

Research Intern, Quantum Intelligence Research Team, Delft University of Technology

Mentor: Dr. ir. Aritra Sarkar, Delft University of Technology, QuTech

- Contributed in the development of a Python package to estimate the **Kolmogorov complexity** of time-evolving Hamiltonians simulated on a quantum processor in terms of program length and circuit depth.
- Derived a general equation to calculate the entropy of a Python program using **Statistical Mechanics** and **Algorithmic Information Theory (AIT)**.
- Developed a method to calculate the complexity of a NAND gate implemented on a Turing Machine.

Test on CI engine using Biodiesel as an alternate fuel.

June 2018 - June 2019

B.E. Research Project.

Mentor: Mr. Rajendra A. Solunke, Assistant Professor, Mechanical Engineering Dept., NBSSOE.

- **Project Aim:** Mapped the performance characteristics of a CI engine by varying its mixture percentage on different loading conditions using biodiesel in combination with petroleum diesel.

Teaching Experience

University of Colorado, Boulder

August 2025 - present (part-time)

Course Assistant - Intro to Batteries, Chemistry for Energy and Mat. Sci., ML in Mech. Engg.

- Doubt resolution and guidance on assignments.
- Grading exams and assignments.

University of Colorado, Boulder

January 2025 - May 2025 (part-time)

Course Assistant - Heat Transfer

- Created regular assignments and their solutions for the students.
- Held weekly office for students for doubt clearance and guidance on assignments.
- Graded exam papers, created rubrics and solutions.

University of Colorado, Boulder

August 2024 - December 2024 (part-time)

Course Assistant - Machine Learning in Mechanical Engineering

- Designed course material for in-class presentation.

Industrial Experience

Merlin Industries, Inc., Pune

November 2021 - April 2024

Design Engineer

- Custom designing about 10 work products like swimming pool liners and covers using AutoCAD daily.
- Helped the team towards higher efficiency by **resolving errors** and being the top performer **in a team of 14**.
- Assisted the team to share load and ensure timely and error-free **delivery of 140 work products daily**.
- Optimized the process using LISPS, reducing the processing time by approximately 50 percent of time.

Mega Process Technology Pvt. Ltd., Pune

January - February - 2018

Internship: Heat Transfer, Refrigeration and Air Conditioning

- Studied the Utility section: Hot Water Generator, Water Treatment Plant and HVAC, and systems in a dairy.

Computing Skills

Programming Languages Python

Frameworks & Libraries IBM Qiskit | PennyLane | Cirq | Jupyter | Matplotlib | Numpy | Pandas

Other Skills Quantum ESPRESSO | $\text{\LaTeX}$ | AutoCAD | ANSYS | CATIA | MATLAB

Conferences

MRS Spring 2026

April 2026

Poster title: Development of a Plasmonic Nanomaterial Platform for Synergistic Thermophotonics

- Presented Poster in the Symposium: EN02: Emergent Materials Paradigms—Catalytic, Photonic and Energy Conversion Technologies for a Sustainable Future

WEFTEC 2025

September 2025

Water Environment Federation's Technical Exhibition and Conference

- Engaged with industry experts, establishing partnerships and securing pilot studies for the experimental validation of the thermo-photocatalytic system.

Rocky Mountain Water Conference

August 2025

Pitch: Thermo-Photocatalysis: Tackling PFAS Synergistically

Innovative Water Technology (IWT) Shark Tank; 1 of 5 Abstracts accepted to present